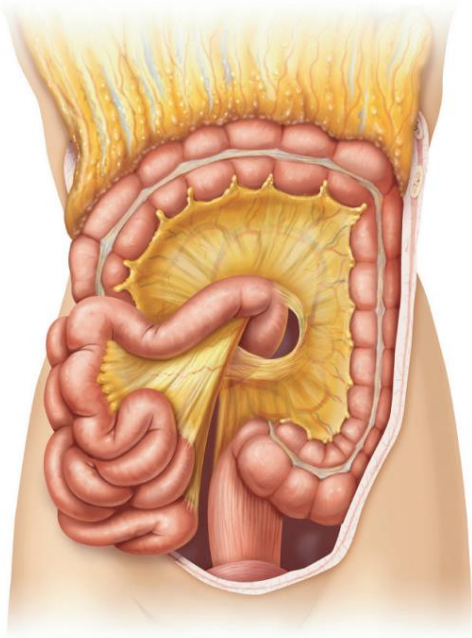
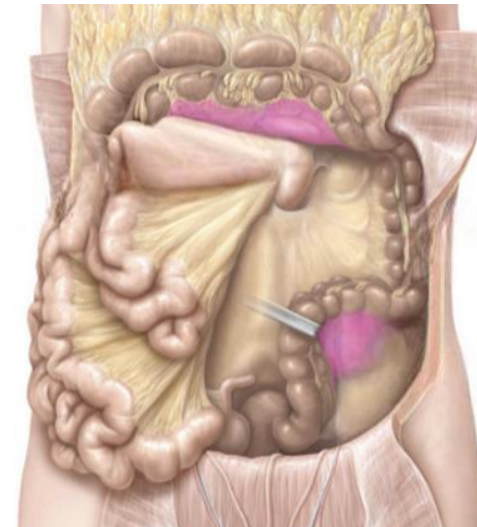
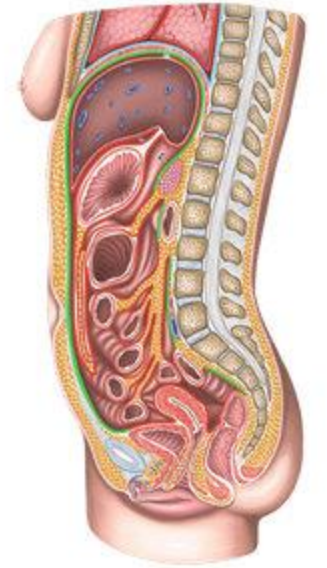


PERITONEUM



By:
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Objectives

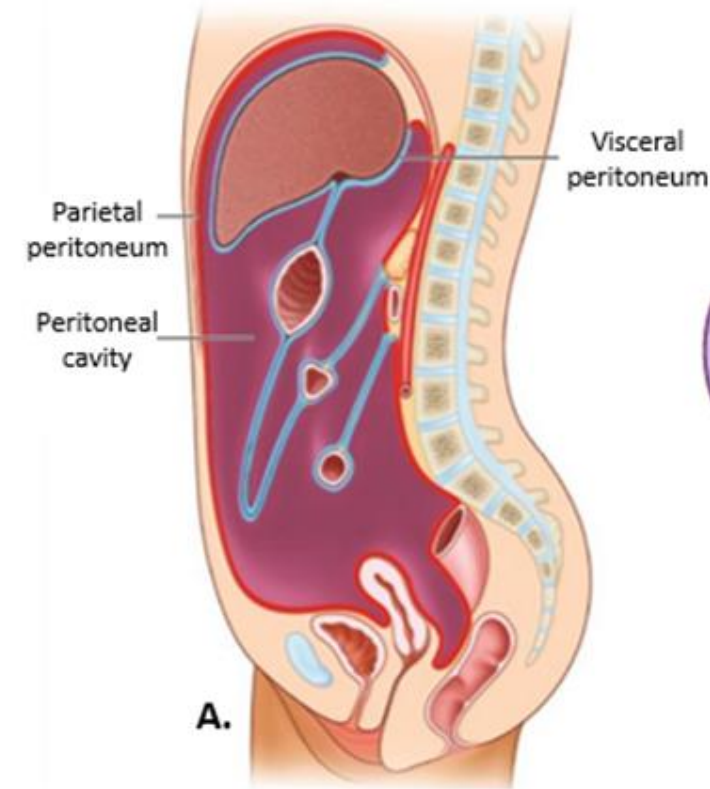
At the end of the lecture the students must know

- ❖ Brief knowledge about the peritoneum as a thin serous membrane and its main parts: parietal and visceral
- ❖ The peritoneal cavity and its parts: the greater sac and the lesser sac (omental bursa)
- ❖ The peritoneal folds: Omenta, Mesenteries, and ligaments
- ❖ The epiploic foramen and its boundaries
- ❖ Mesentery of the small intestine and ligaments of the liver
- ❖ Nerve supply of the peritoneum.
- ❖ Clinical points.

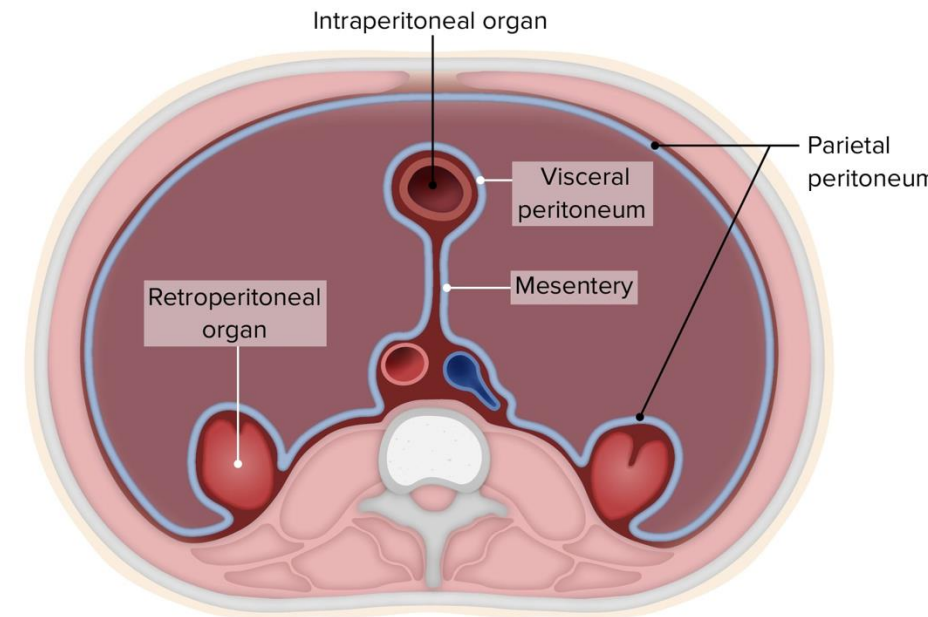
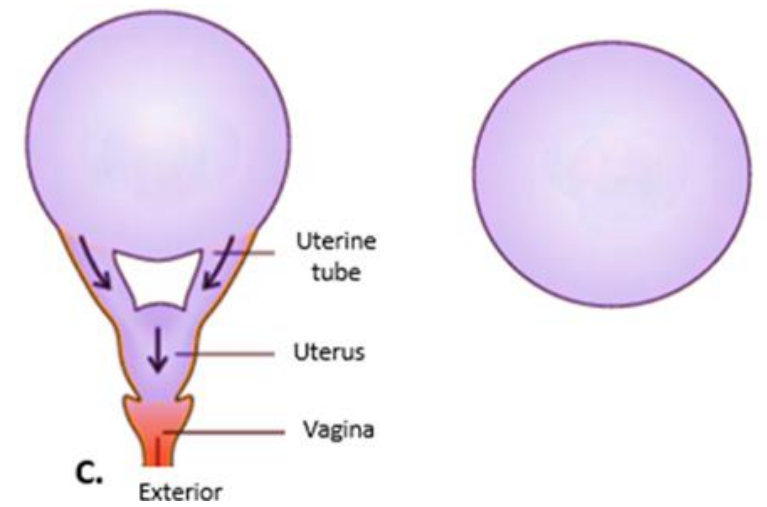


PERITONEUM

- ❑ The peritoneum is a large, thin serous membrane, which lines the interior of the abdominopelvic cavity
- ❑ It has a similar structure to the pleura and serous pericardium, consisting of parietal and visceral layers.
- ❑ Parietal peritoneum: Lines the internal surface of the abdominal and pelvic walls and is supplied by the same blood vessels as the overlying abdominal wall. It is sensitive to pain, pressure, and temperature changes
- ❑ Visceral peritoneum: It covers the external surfaces of most abdominal organs.
- ❑ Peritoneal cavity: filled with a thin film of fluid (50 mL in adults), that lubricates and facilitates movement of abdominal viscera

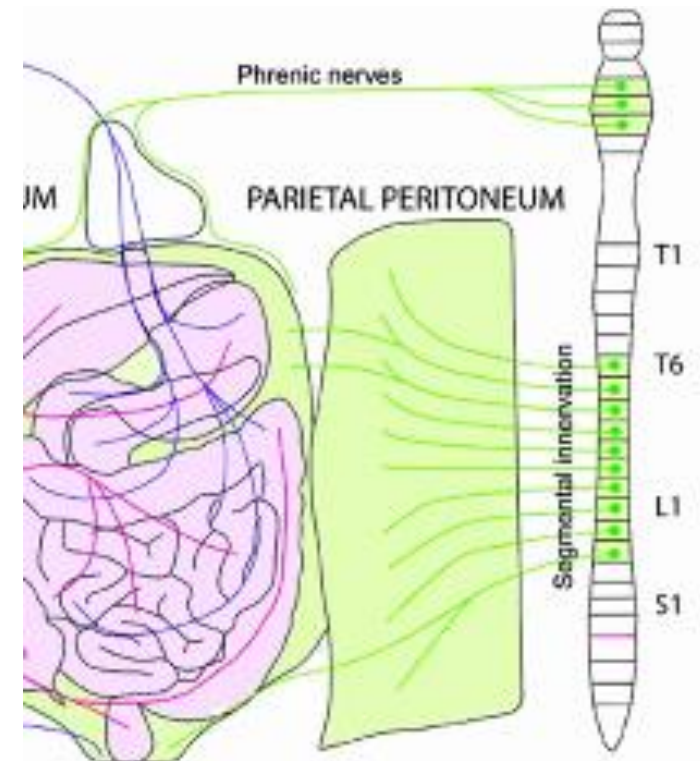


- ❑ In males, the peritoneum is a closed serous sac
- ❑ In females, it is not a closed sac because it communicates with the exterior through the uterine tubes, uterus, and vagina
- ❑ Some organs lie on the posterior abdominal wall and are only partially covered by peritoneum on their anterior surface; these are referred to as retroperitoneal organs.
- ❑ Other organs are completely surrounded by peritoneum on all sides; these are termed intraperitoneal organs.
- ❑ Examples of retroperitoneal organs include: Kidneys, Suprarenal glands, Ureters, Pancreas, Duodenum, Ascending colon and Descending colon
- ❑ Examples of intraperitoneal organs include: Stomach, Spleen, Liver, Transverse colon, Sigmoid colon, Jejunum and Ileum



Nerve Supply of the Peritoneum

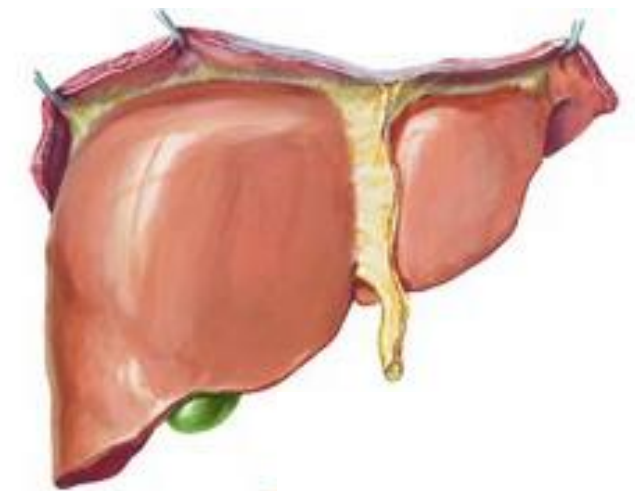
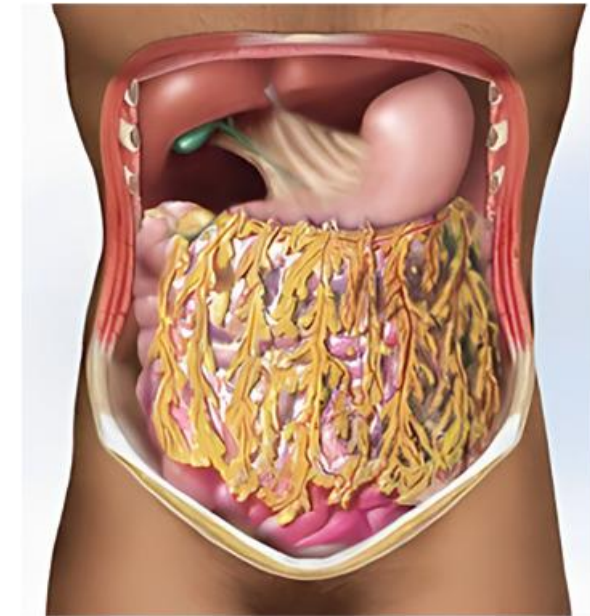
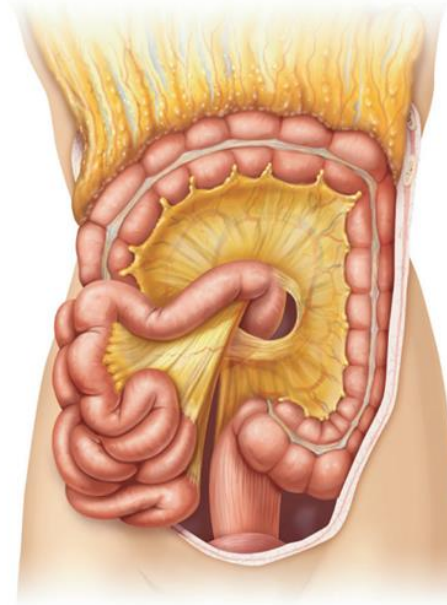
- ❑ The parietal peritoneum: is sensitive to pain, touch, pressure, and temperature.
- ❑ The parietal peritoneum of the anterolateral abdominal wall is supplied by the T7 to T12 and L1 spinal nerves, like the skin, muscles, and fascia of the abdominal wall.
- ❑ The central part of the diaphragmatic peritoneum is supplied by the phrenic nerve.
- ❑ The peripheral part of the diaphragmatic peritoneum is innervated by the T7 to T12 and L1 spinal nerves.
- ❑ The peripheral part of the pelvic peritoneum is supplied by the obturator nerve.
- ❑ Visceral Peritoneum: is sensitive only to stretching and tearing and is innervated by autonomic afferent nerves that supply the organs.



Classification of the peritoneal folds

The peritoneal folds are classified into three types: mesentery/mesocolon, omenta, and ligaments.

1. **Mesentery/mesocolon:** The fold that connects the intestine to the posterior abdominal wall.
2. **Omenta (singular: omentum):** These are the peritoneal folds that connect the stomach with other organs
3. **Ligaments:** These are condensed folds of peritoneum that connect organs to the abdominal wall or to each other.

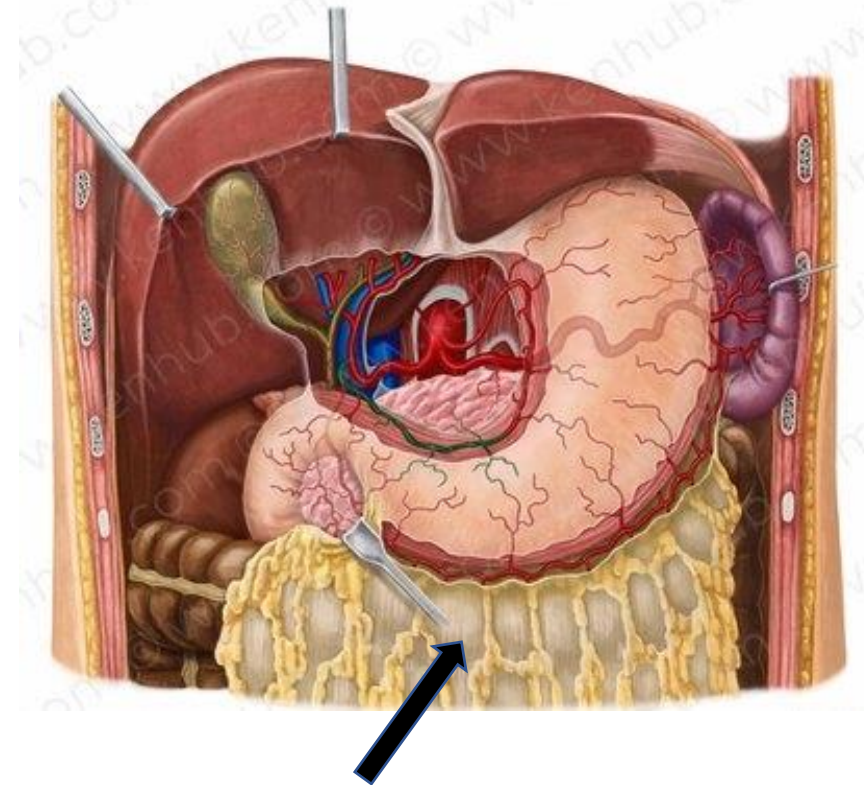


Greater Omentum (L. Omentum = Apron)

The greater omentum is a large, thick fold of peritoneum extends from the greater curvature of the stomach and one inch of the duodenum to the transverse colon and transverse mesocolon.

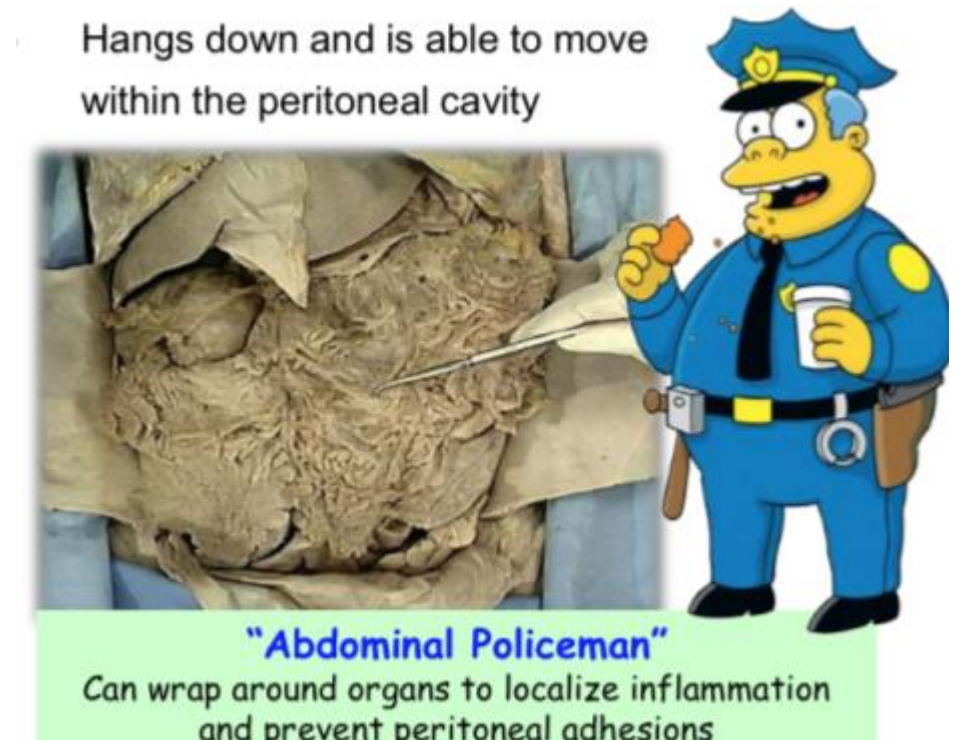
Contents

- Adipose tissue (fat) in varying amounts dependent on the individual's nutritional status.
- Aggregations of macrophages that form dense patches known as milky spots.
- Right and left gastroepiploic vessels.



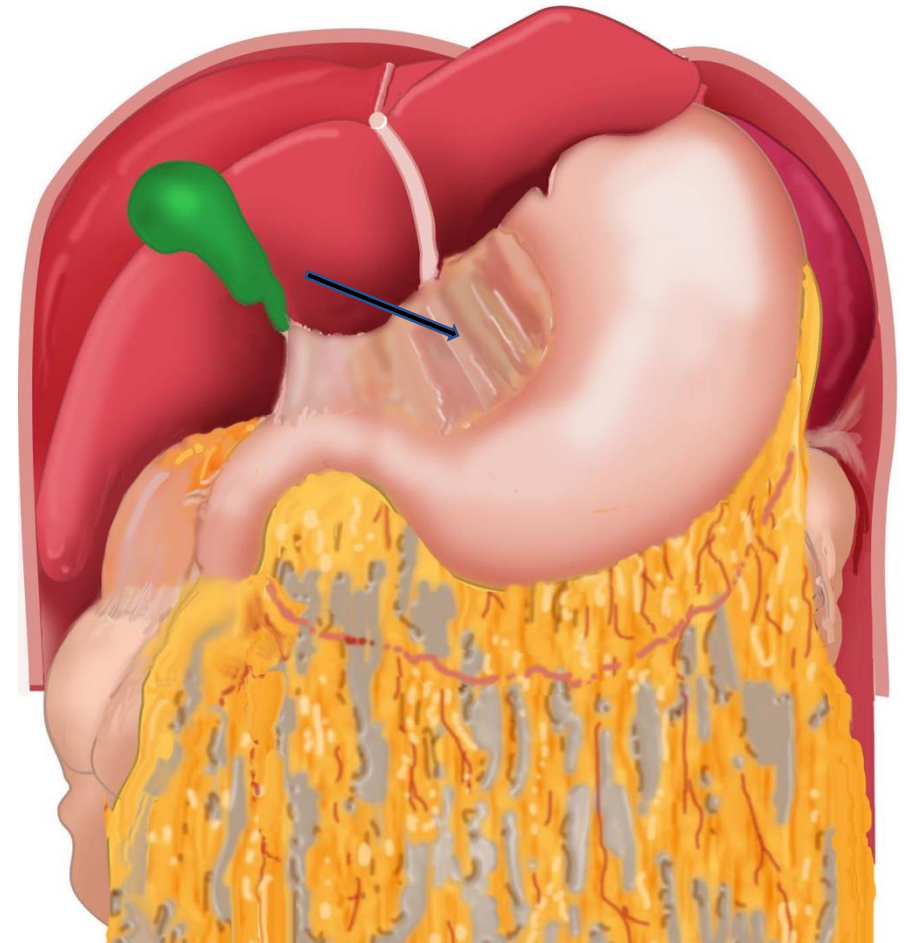
CLINICAL ANATOMY

- ❑ The greater omentum protects the peritoneal cavity from infection due to the presence of a large number of macrophages.
 - ❑ It limits the spread of infection by moving to the site of infection and sealing it off from surrounding areas.
 - ❑ For this reason, the greater omentum is often referred to as the "policeman of the abdomen."
- Additionally, it serves as a storehouse for adipose tissue and is sometimes used as grafting material by surgeons.



Lesser Omentum

- ❑ The lesser omentum is a double-layered fold of peritoneum that extends between the lesser curvature of the stomach and the inferior surface of the liver.
- ❑ Inferiorly, it attaches to the right side of the abdominal esophagus, the lesser curvature of the stomach, and the first 2 cm of the duodenum.
- ❑ Superiorly, it connects in an inverted L-shaped manner to the margins of the fissures for the ligamentum venosum and the porta hepatis.



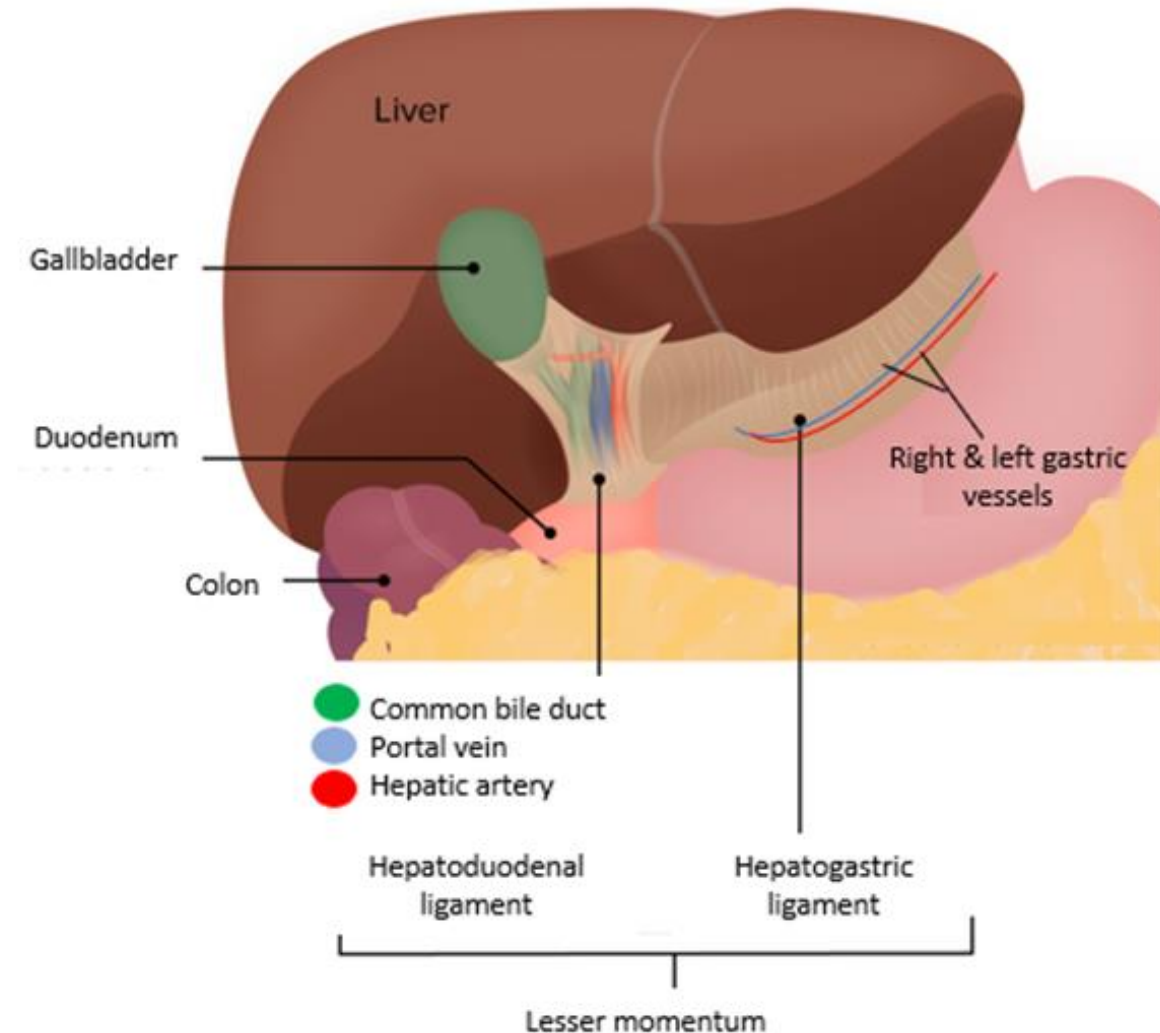
Contents of the lesser Omentum

□ In the free margin (hepatoduodenal ligament) superficial → deep:

- Bile duct (right side, anterior)
- Hepatic artery proper (left side, anterior)
- Portal vein (posterior, deepest)

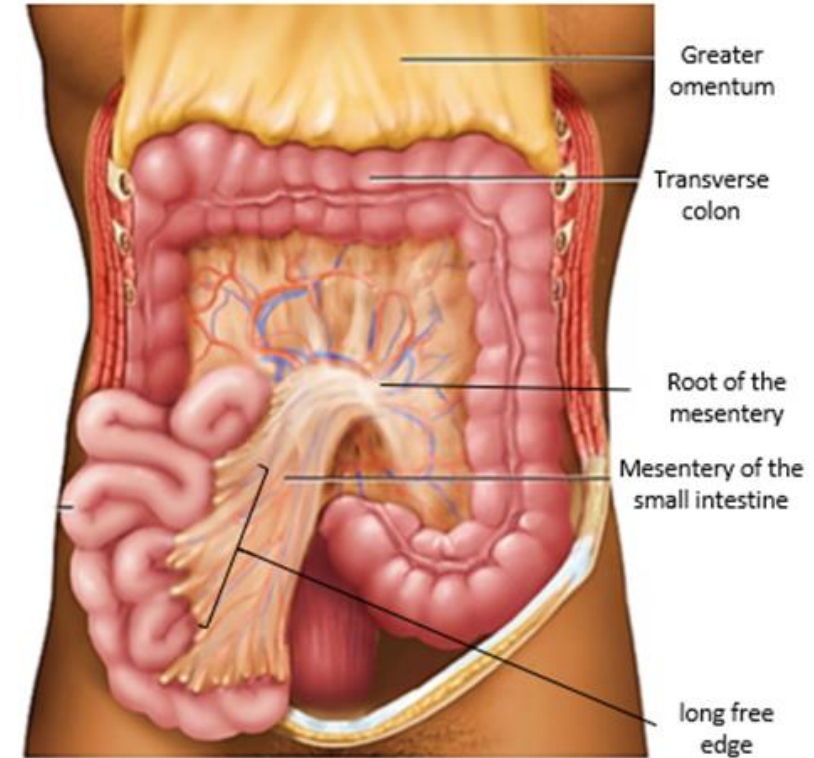
□ In the hepatogastric ligament (superficial → deep):

- Right gastric vessels
- Left gastric vessels
- Lymphatics & lymph nodes
- Branches of vagus nerves
- Sympathetic fibers from celiac plexus



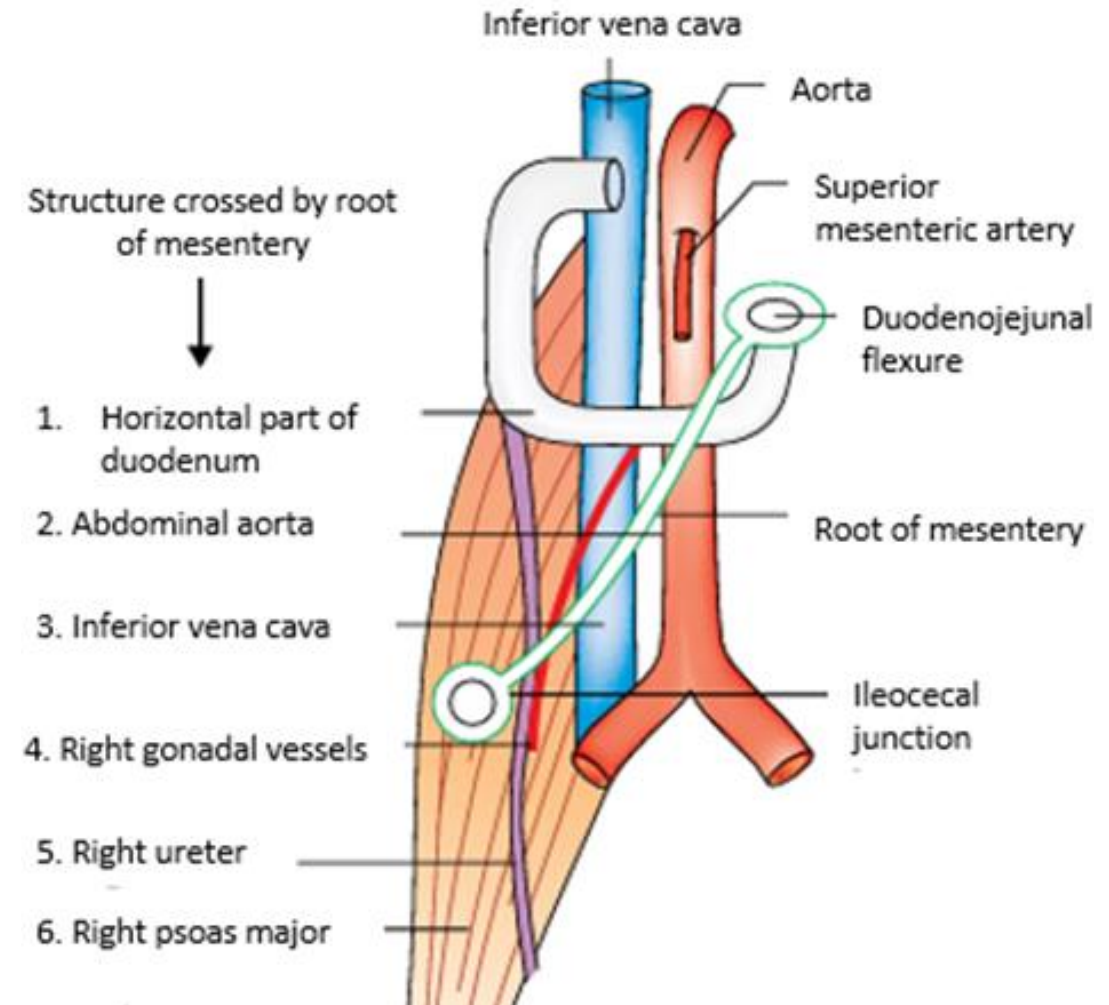
Mesentery (Mesentery of Small Intestine)

- ❑ It is a broad fan-shaped fold of the peritoneum that suspends the coils of the small intestine (jejunum and ileum) from the posterior abdominal wall. It has two borders: the attached border (or root) and long free edge (intestine border)
- ❑ the root divides the infracolic compartment into two parts: right and left
- ❑ **Free border (intestinal border):** It is about 6 m (20 feet) long and encloses the jejunum and ileum and accounts for the formation of folds (pleats) in it (a frill-like arrangement). Its length permits the free mobility of the loops of jejunum and ileum in the abdominal cavity.



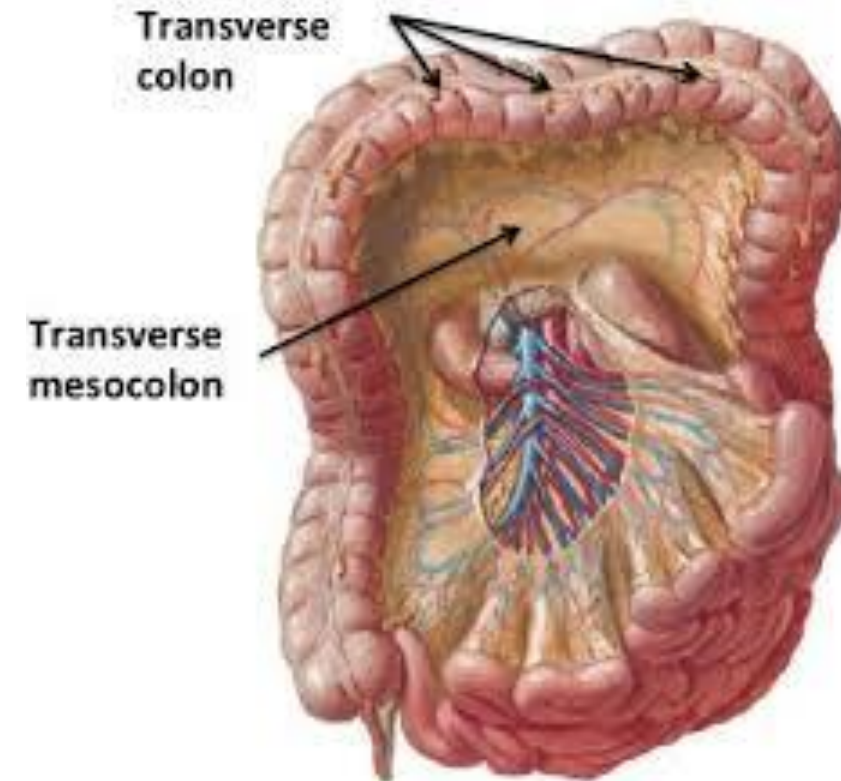
□ Root of the mesentery

- It is attached to an oblique line across the posterior abdominal wall, extending from the duodenojejunal flexure (left side of L2) to the ileocecal junction (right sacroiliac joint)
- It is crossed from above downward by the horizontal (third) part of the duodenum, abdominal aorta, inferior vena cava, right gonadal vessels, right ureter, and right psoas major muscle.
- The mesentery contains the superior mesenteric artery, vein, and lymphatic vessels, as well as nodes, fat, and nerve fibers.



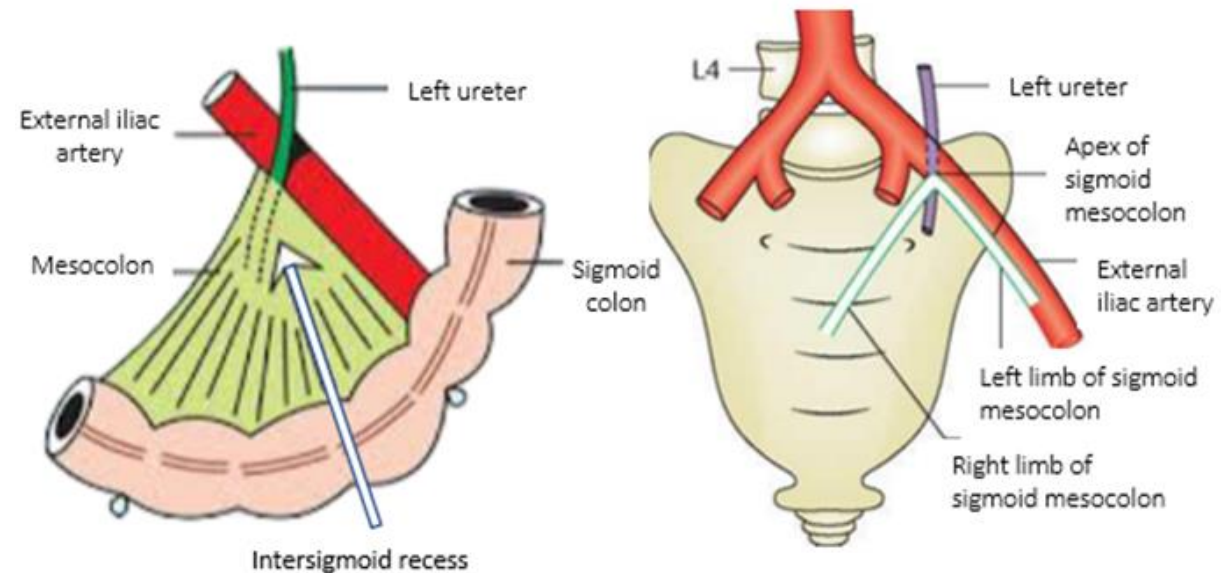
Transverse Mesocolon

- ❑ The transverse colon is suspended from the posterior abdominal wall by the transverse mesocolon, a large double-layered fold of peritoneum.
- ❑ It is fused with the posterior surface of the greater omentum and divides the peritoneal cavity into supra- and infracolic compartments.
- ❑ It acts as a natural barrier against the spread of infection between these compartments.
- ❑ The contents of the transverse mesocolon include the middle colic vessels, ascending branches of the right and left colic vessels, lymph nodes, lymphatics, and nerve plexuses and fat



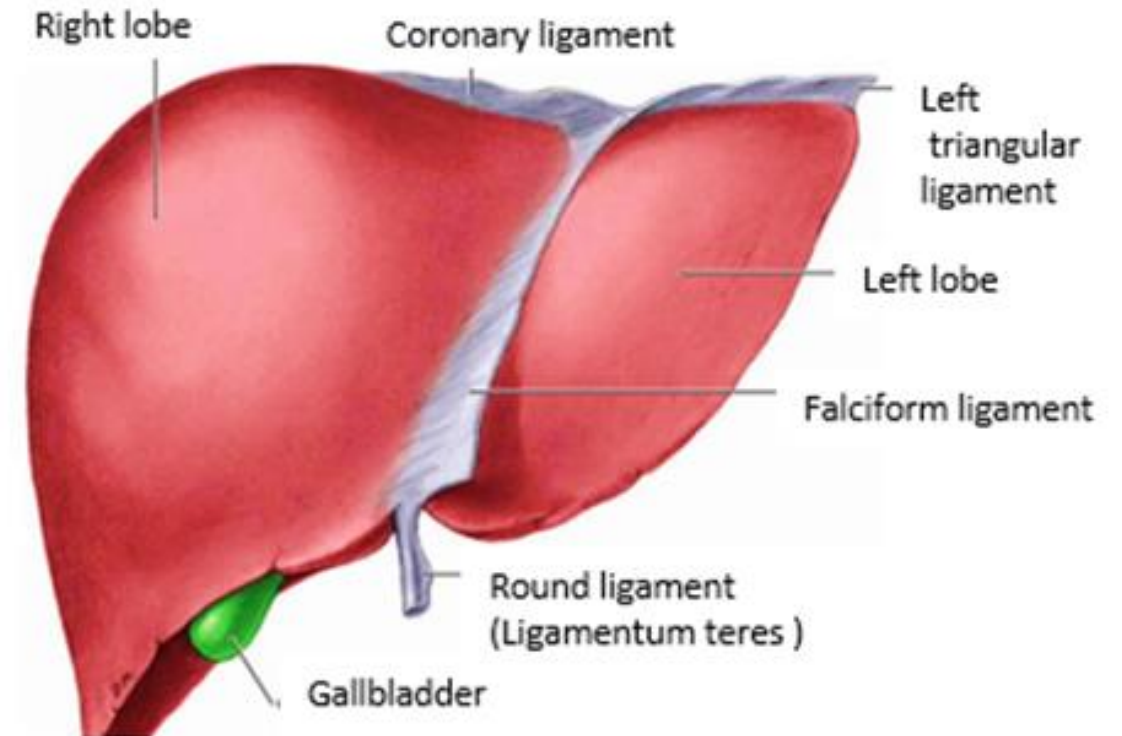
Sigmoid (Pelvic) Mesocolon

- ❑ The sigmoid colon is suspended from the pelvic wall by a large peritoneal fold known as the sigmoid mesocolon, which has an inverted V-shaped attachment or root
- ❑ The root consists of left and right limbs, along with an apex.
- ❑ Left limb connects to the external iliac artery
- ❑ Right limb is attached to the pelvic surface of the sacrum.
- ❑ The meeting point between the right and left limbs is called the apex. The point where the left ureter and left common iliac vessels cross each other.



Falciform Ligament

- ❑ The falciform ligament is a sickle-shaped fold of peritoneum, which connects the anterosuperior surface of the liver to the supraumbilical part of the anterior abdominal wall and the inferior surface of the diaphragm

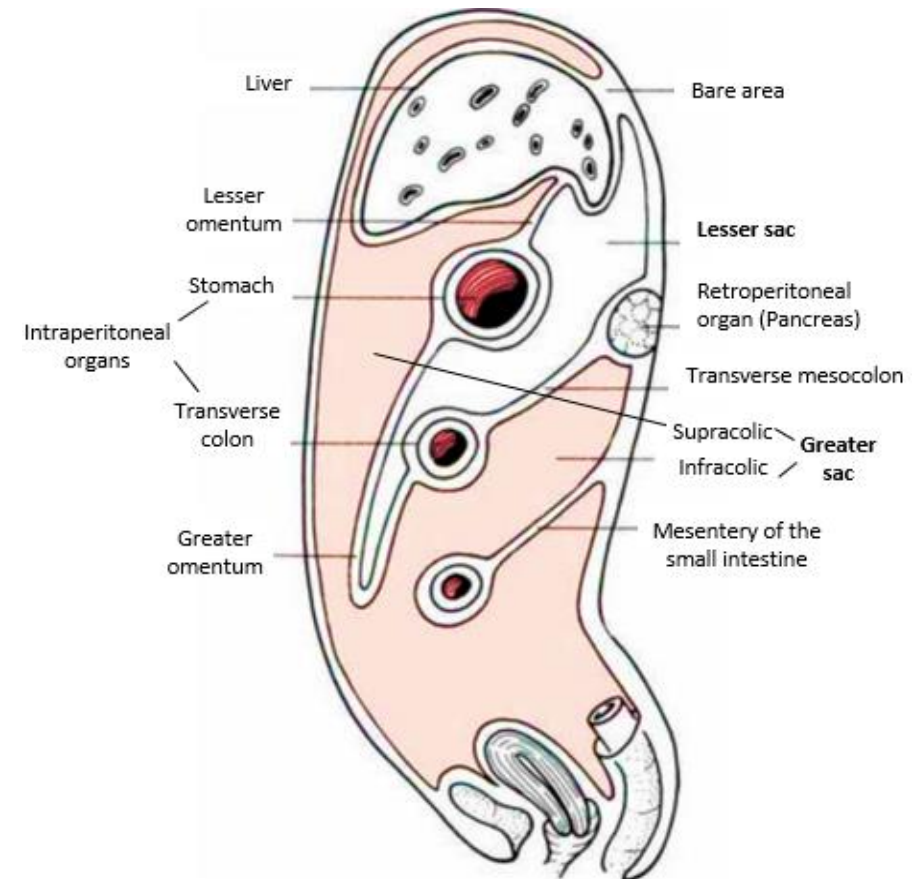


PERITONEAL CAVITY

- The peritoneal cavity is the largest and most complex serous sac in the body, located between the parietal and visceral layers of the peritoneum.
- In males, it is a closed cavity; however, in females, it communicates with the exterior through the ostia of the uterine tubes, uterus, and vagina. Consequently, pelvic infections are more common in females.
- The peritoneal cavity divided into two parts; they communicate with each other through the epiploic foramen.

1. Greater sac

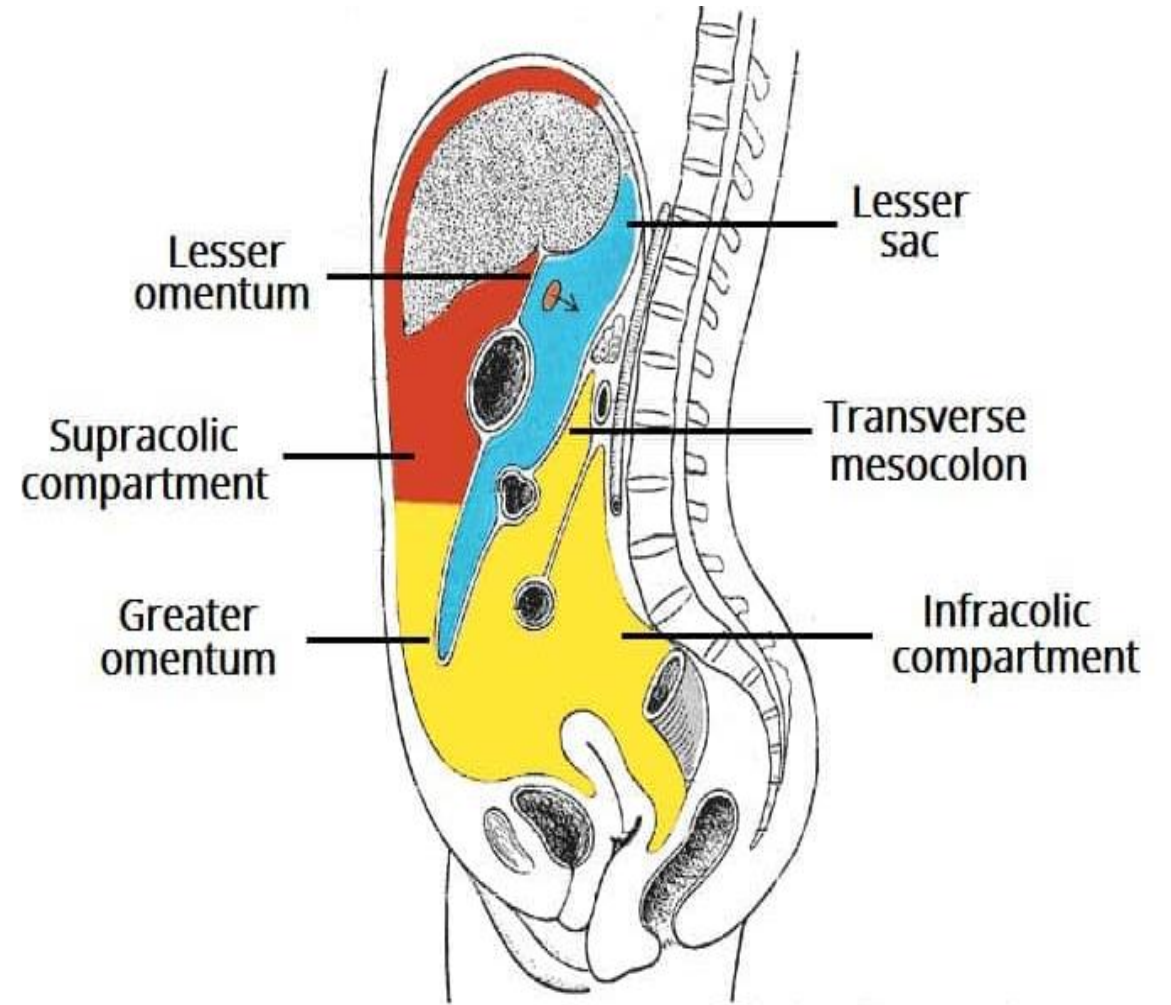
2. Lesser sac (or omental bursa)



The greater sac

This is the larger (main) compartment of the peritoneal cavity; it is bounded by:

- ❑ **Superior:** diaphragm
- ❑ **Inferior:** extends into the pelvic cavity
- ❑ **Anteriorly:** anterior abdominal wall
- ❑ **Posteriorly:** posterior abdominal wall, including the lumbar vertebrae, psoas major, and quadratus lumborum.

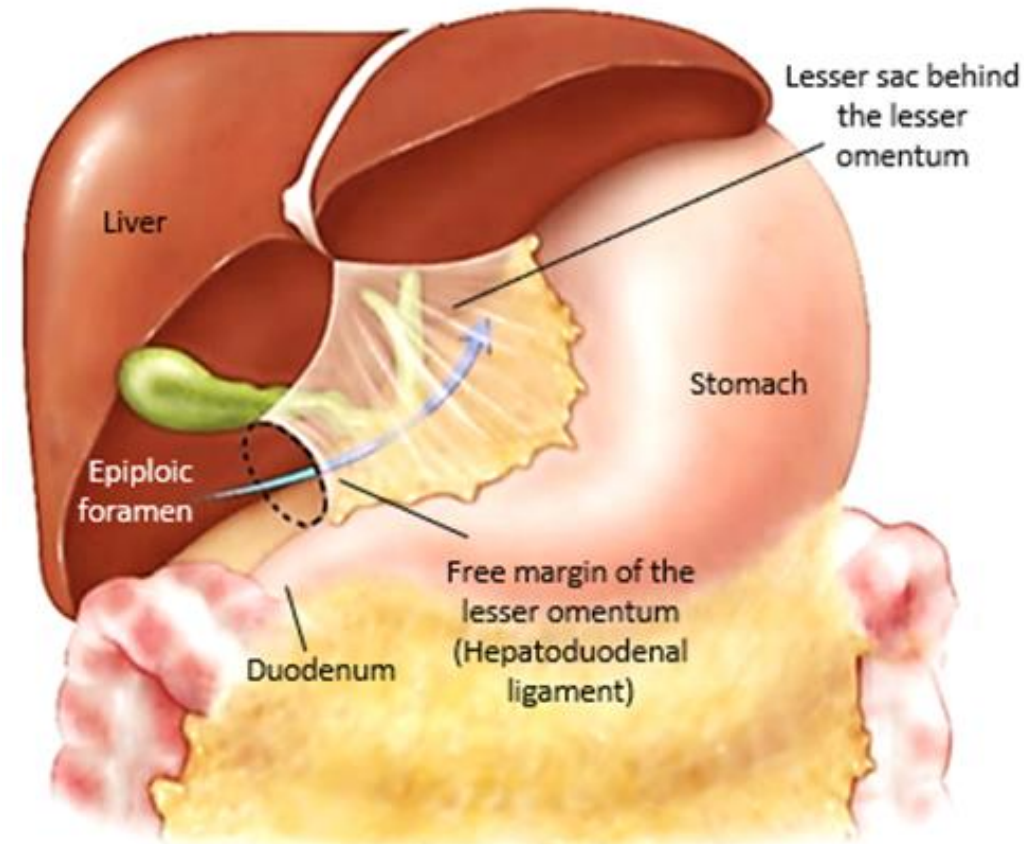


The lesser sac

❑ This smaller compartment lies behind the stomach, liver, and lesser omentum

❑ It is bounded by:

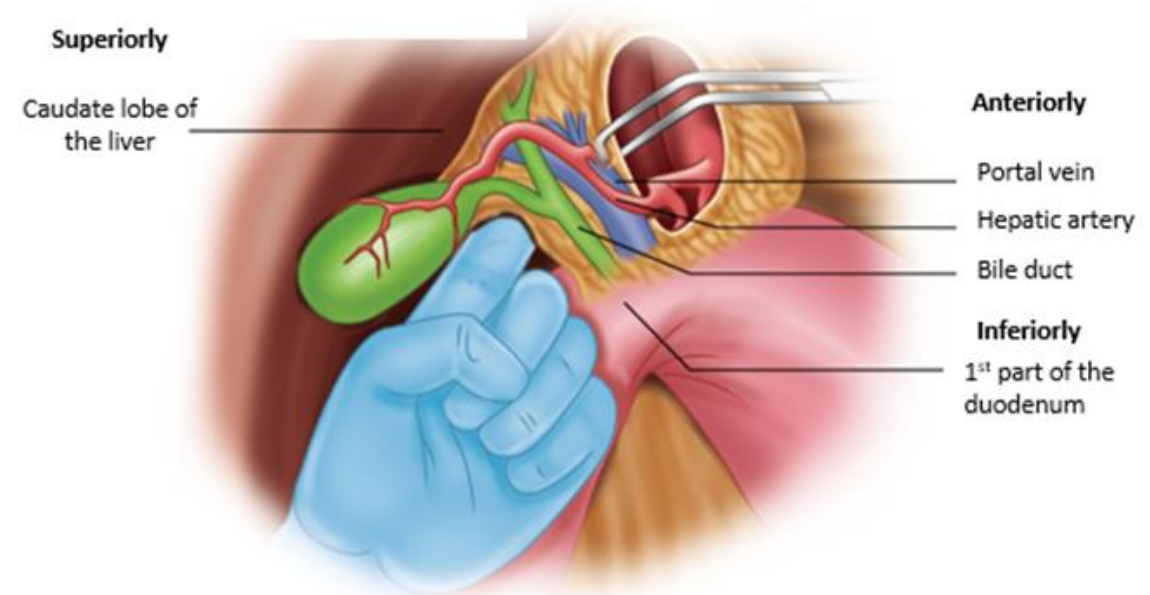
- **Anteriorly:** Lesser omentum and stomach,
- **Posteriorly:** Pancreas, left kidney, left adrenal gland, aorta, and inferior vena cava
- **Superiorly:** Caudate lobe of the liver and diaphragm
- **Inferiorly:** Transverse mesocolon greater omentum
- **Left:** Gastrosplenic ligament and splenorenal ligament
- **Right:** Epiploic foramen



Foramen epiploicum (Omental foramen)

❑ The two sacs communicate with each other through the foramen epiploicum, has the following boundaries:

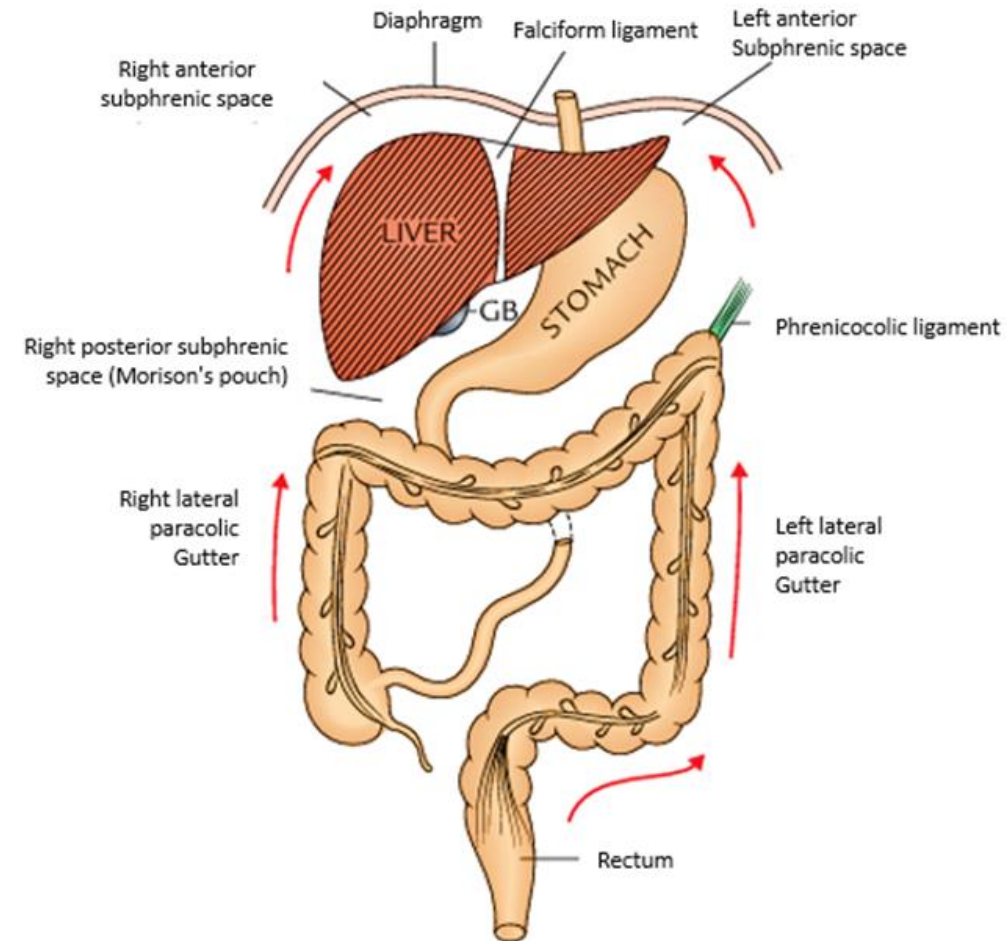
- **Anteriorly:** hepatoduodenal ligament (containing the portal triad: portal vein, hepatic artery, bile duct)
- **Posterior:** inferior vena cava
- **Superior:** caudate lobe of the liver
- **Inferior:** first inch of the first part of the duodenum



Peritoneal Pouches, Recesses, Spaces, and Gutters

❑ Peritoneal pouches, recesses, spaces, and gutters are possible areas in the abdominal cavity that are made by the folds of the peritoneum. These areas are clinically significant because they can collect fluids like blood, pus, or ascitic fluid. They could also be places where infections spread and where internal hernias happen.

❑ The abdominal part of the peritoneal cavity is divided into supra-colic and infra-colic compartments by the transverse colon and its mesentery

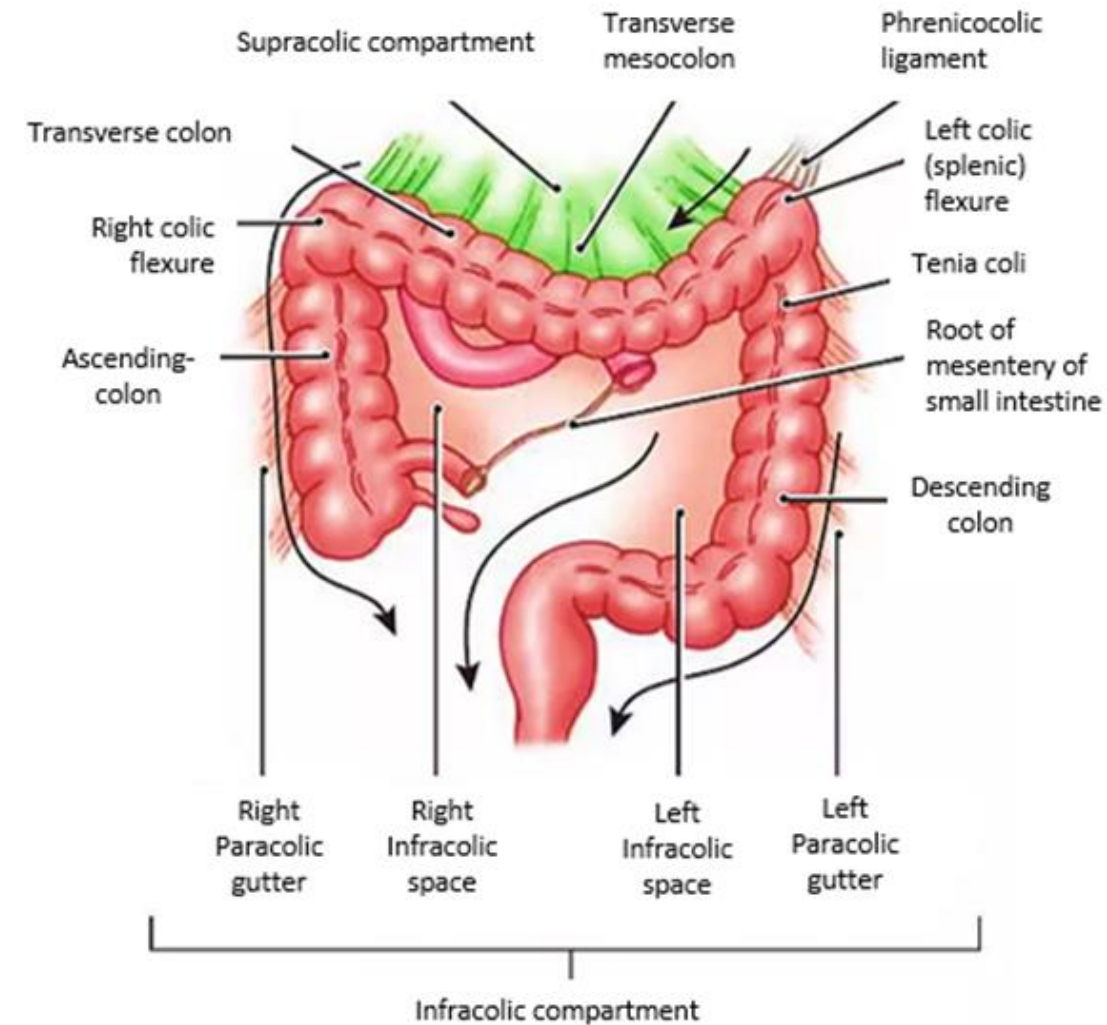


❑ **Supra-colic Compartment:** Is primarily located beneath the costal margin and diaphragm. It contains the liver, stomach, spleen, and the superior part of the duodenum.

❑ **Infra-colic Compartment:** Is filled with the coils of the jejunum and ileum and is surrounded by the ascending, transverse, and descending colons.

❑ It is divided into right and left infracolic spaces by the root of the mesentery of the small intestine

- The right paracolic gutter connects the hepatorenal pouch with the pelvis.
- The left paracolic gutter is limited above by the phrenicocolic ligament.



❑ Subphrenic Spaces

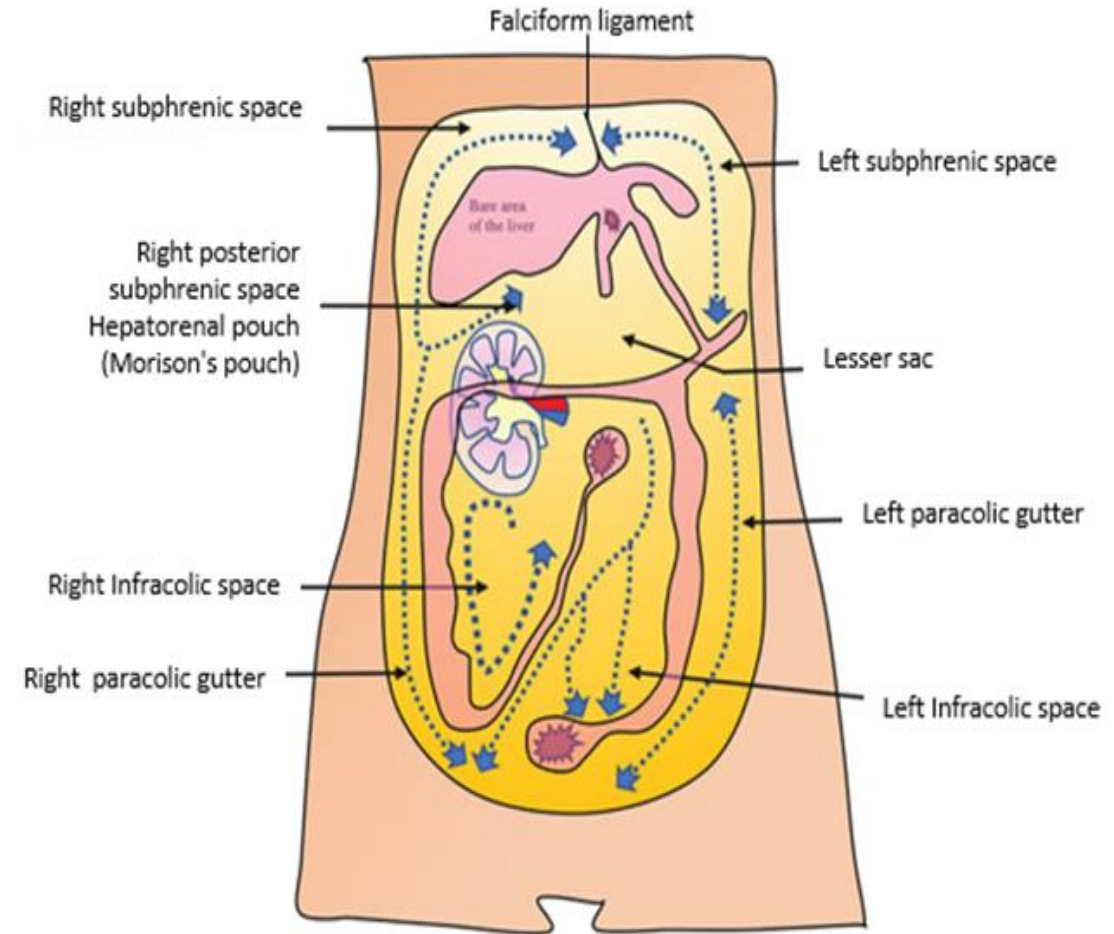
○ The subphrenic space is the space located below the diaphragm and above the liver, divided into right and left compartments by the falciform ligament.

❑ **Hepatorenal Pouch or Morrison's Pouch** is located between the posteroinferior surface of the liver and the front of the right kidney

○ It is clinically significant as the lowest peritoneal recess in the supine position, making it an important site for diagnosing and detecting free fluid, trauma-related hemorrhage, abscesses, and ascites.

○ Most intraperitoneal spaces connect to this pouch,

so even small amounts of bleeding or extra fluid in the peritoneum will drain into it.

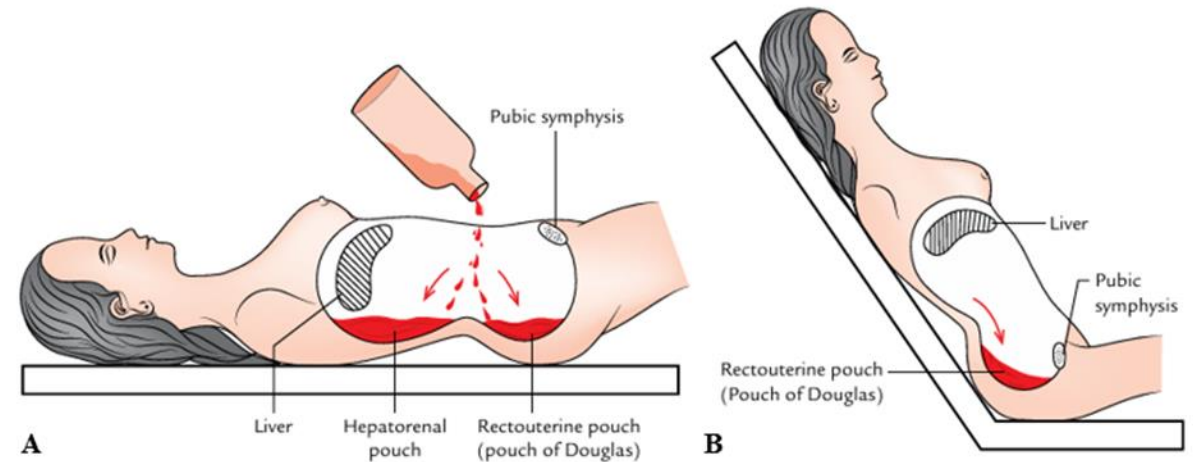


Clinical significance

- ❑ The hepatorenal pouch is the most dependent part of the abdominal cavity when a person is in the supine position

- ❑ located above the pelvic brim. Consequently, pus can accumulate in this area following upper abdominal surgeries, such as cholecystectomy.

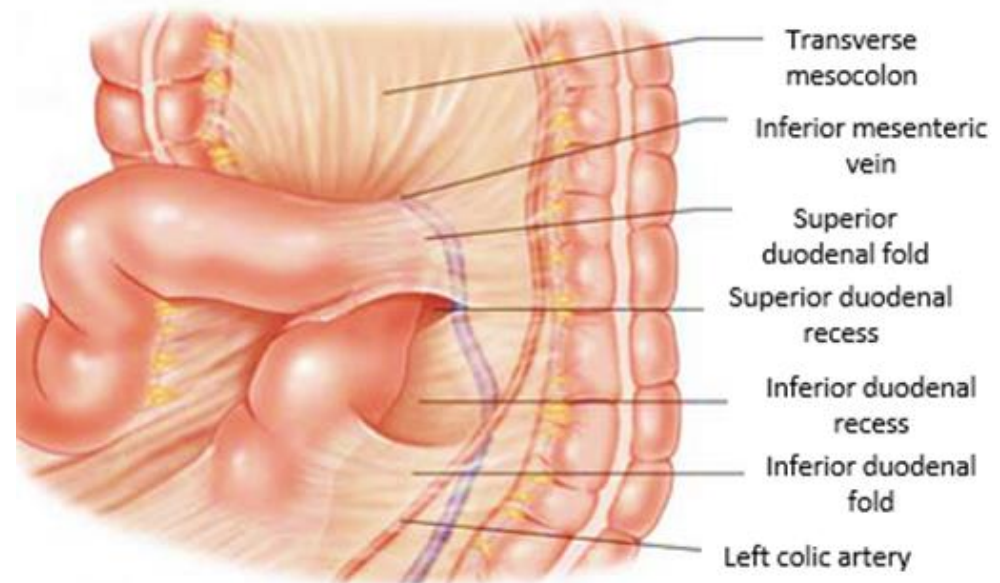
- ❑ Therefore, it is standard practice for surgeons to insert a tube into this pouch to enable gravitational drainage.



- ❑ Additionally, the hepatorenal pouch and the rectouterine pouch are interconnected through the right paracolic gutter, which serves as a pathway for infection to spread from the abdomen to the pelvis.

Duodenal recesses (fossae)

- ❑ In the region of the duodenojejunal junction, small pocket-like pouches of peritoneum called duodenal recesses
- ❑ These are sometimes responsible for strangulated retroperitoneal hernia.
- ❑ The important peritoneal recesses around the duodenojejunal flexure are as follows:
 - Superior duodenal recess
 - Inferior duodenal recess
 - Paraduodenal recess
 - Retroduodenal recess





CLINICAL

CASE

SCENARIO



A patient develops an abscess beneath the diaphragm after upper abdominal surgery. Imaging shows two compartments separated by the falciform ligament. **Which spaces are these?**

- a) Right and left subphrenic spaces
- b) Hepatorenal and subhepatic spaces
- c) Rectouterine and vesicouterine pouches
- d) Supra- and infracolic spaces
- e) Omental bursa compartments



CLINICAL

CASE

SCENARIO



- A male patient has peritoneal fluid accumulation between the rectum and the urinary bladder.

Which peritoneal pouch is this?

- A. Rectouterine pouch
- B. Rectovesical pouch
- C. Hepatorenal pouch
- D. Vesicouterine pouch
- E. Subphrenic pouch



CLINICAL

CASE

SCENARIO



A patient with perforated appendix develops pus that tracks along the lateral and medial sides of the ascending and descending colon.

Which structures allow this spread?

- A. Subphrenic spaces
- B. Paracolic gutters
- C. Hepatorenal pouch
- D. Lesser sac
- E. Sigmoid mesocolon



CLINICAL

CASE

SCENARIO



A trauma patient has fluid accumulation in the upper peritoneal compartment. Imaging shows liver, stomach, spleen, and superior duodenum involvement.

Which compartment is this?

- A. Infra-colic compartment
- B. Supra-colic compartment
- C. Lesser sac
- D. Pelvic cavity
- E. Omental bursa



CLINICAL

CASE

SCENARIO



A patient has an internal hernia where a loop of small intestine passes from the greater sac into the lesser sac. Through which structure did the intestine herniate?

- A. Transverse mesocolon
- B. Epiploic (omental) foramen
- C. Falciform ligament
- D. Hepatogastric ligament
- E. Sigmoid mesocolon

Thank You